

1. Create a console-based app called Music Listening Logger that allows users to enter the number of minutes they listened to music each day for a week and stores this data in an array.

```
let musicMinutes = [];
```

```
console.log("Music Listening Logger");
```

```
for (let i = 0; i < 7; i++) {  
    let minutes = Number(prompt("Enter listening minutes for day " + (i + 1) + ":"));  
    musicMinutes.push(minutes);  
}
```

```
console.log("Weekly Listening Data:");
```

```
console.log(musicMinutes);
```

**2. Add a menu-driven interface to your Music Listening Logger so users can choose to log new listening minutes, view their weekly summary, or exit the app.

Hint: Use a loop to repeatedly show the menu until the user selects exit.**

```
let musicMinutes = [];
```

```
let choice;
```

```
do {  
    console.log("\n--- Music Listening Logger ---");  
    console.log("1. Log listening minutes");  
    console.log("2. View weekly summary");  
    console.log("3. Exit");
```

```
choice = Number(prompt("Enter your choice:"));

if (choice === 1) {

    let day = Number(prompt("Enter day number (1-7):"));
    let minutes = Number(prompt("Enter listening minutes:"));

    musicMinutes[day - 1] = minutes;

    console.log("Listening minutes saved.");

} else if (choice === 2) {

    console.log("Weekly Listening Data:");

    for (let i = 0; i < 7; i++) {
        console.log("Day " + (i + 1) + ": " + (musicMinutes[i] || 0) + " minutes");
    }

} else if (choice === 3) {

    console.log("Thank you for using Music Listening Logger.");

} else {

    console.log("Invalid choice.");
```

```
}
```

```
} while (choice !== 3);
```

3. Implement file handling in your Music Listening Logger so that when the user logs their daily listening minutes, the data is saved to a file named music_log.txt for persistence.

For file handling, Node.js provides the fs module.

```
const fs = require("fs");
```

```
let musicMinutes = [];
```

```
console.log("--- Music Listening Logger ---");
```

```
for (let i = 0; i < 7; i++) {
```

```
    let minutes = Number(
```

```
        prompt("Enter listening minutes for day " + (i + 1) + ":")
```

```
    );
```

```
    musicMinutes.push(minutes);
```

```
}
```

```
// Convert array into text
```

```
let data = musicMinutes.join("\n");
```

```
// Save data into file
```

```
fs.writeFileSync("music_log.txt", data);
```

```
console.log("Music data saved to music_log.txt");
```

The file will contain something like:

60

45

90

30

120

75

50

4. Add a feature to your app that reads the saved music_log.txt file and generates a weekly report showing the total, average, and highest listening minutes in the week.

```
const fs = require("fs");
```

```
// Read data from the file
```

```
let data = fs.readFileSync("music_log.txt", "utf8");
```

```
// Convert file data into an array of numbers
```

```
let musicMinutes = data.trim().split("\n").map(Number);
```

```
let total = 0;
```

```
let highest = musicMinutes[0];
```

```
// Calculate total and highest
```

```
for (let i = 0; i < musicMinutes.length; i++) {
```

```
total = total + musicMinutes[i];

if (musicMinutes[i] > highest) {
    highest = musicMinutes[i];
}
}

// Calculate average
let average = total / musicMinutes.length;

console.log("\n--- Weekly Music Report ---");
console.log("Total Minutes: " + total);
console.log("Average Minutes: " + average);
console.log("Highest Minutes: " + highest);
```

For example, if the file contains:

```
60
45
90
30
120
75
50
```

The report will be:

```
--- Weekly Music Report ---
```

```
Total Minutes: 470
```

```
Average Minutes: 67.14285714285714
```

```
Highest Minutes: 120
```

You can make the average cleaner with:

```
console.log("Average Minutes: " + average.toFixed(2));
```

**5. Refactor your Music Listening Logger to allow users to reset their weekly data by clearing the array and deleting the contents of music_log.txt.

Constraint: Make sure the reset option is available in the menu and asks for confirmation before deleting data.**

This version combines **array + menu + file handling + weekly report + reset confirmation.**

```
const fs = require("fs");
```

```
let musicMinutes = [];
```

```
// Load existing data from file if it exists
```

```
if (fs.existsSync("music_log.txt")) {
```

```
    let data = fs.readFileSync("music_log.txt", "utf8");
```

```
    if (data.trim() !== "") {
```

```
        musicMinutes = data.trim().split("\n").map(Number);
```

```
    }
```

```
}
```

```
let choice;
```

```
do {
```

```
    console.log("\n=====");
```

```
    console.log(" MUSIC LISTENING LOGGER");
```

```
console.log("=====");
console.log("1. Log listening minutes");
console.log("2. View weekly report");
console.log("3. Reset weekly data");
console.log("4. Exit");

choice = Number(prompt("Enter your choice: "));

// Option 1: Enter listening minutes
if (choice === 1) {

    let day = Number(prompt("Enter day number (1-7):"));
    let minutes = Number(prompt("Enter listening minutes:"));

    musicMinutes[day - 1] = minutes;

    // Save updated data to file
    fs.writeFileSync(
        "music_log.txt",
        musicMinutes.map(value => value || 0).join("\n")
    );

    console.log("Data saved successfully.");

}

// Option 2: Show weekly report
else if (choice === 2) {
```

```
if (musicMinutes.length === 0) {  
    console.log("No music data available.");  
}  
else {  
  
    let total = 0;  
    let highest = musicMinutes[0];  
  
    for (let i = 0; i < musicMinutes.length; i++) {  
  
        let minutes = musicMinutes[i] || 0;  
  
        total = total + minutes;  
  
        if (minutes > highest) {  
            highest = minutes;  
        }  
    }  
  
    let average = total / 7;  
  
    console.log("\n--- Weekly Report ---");  
    console.log("Total Minutes: " + total);  
    console.log("Average Minutes: " + average.toFixed(2));  
    console.log("Highest Minutes: " + highest);  
}  
}
```

```
// Option 3: Reset data

else if (choice === 3) {

    let confirmReset = prompt(
        "Are you sure you want to delete all weekly data? (yes/no)"
    );

    if (confirmReset.toLowerCase() === "yes") {

        // Clear the array
        musicMinutes = [];

        // Clear the file
        fs.writeFileSync("music_log.txt", "");

        console.log("Weekly data has been reset.");

    } else {
        console.log("Reset cancelled.");
    }
}

// Option 4: Exit
else if (choice === 4) {

    console.log("Thank you for using Music Listening Logger!");

}
```

```
else {  
    console.log("Invalid choice. Please try again.");  
}
```

```
} while (choice !== 4);
```

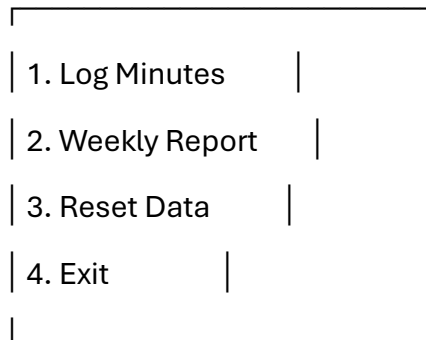
Simple flow

Start

↓

Show Menu

↓



↓

User Choice

↓

Perform Action

↓

Show Menu Again

↓

Exit?

↩ ↘

No Yes

↓ ↓

Menu End